

EE5203 Microprocessors

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Course materials for the 2026-2027 offering: theory slides, study sheets, practice worksheets, and laboratory guides for each lecture.

Lectures 1-4 form a cumulative Embedded C foundation, and each laboratory builds on the previous week's working project. Lecture numbers describe instructional order, not calendar weeks — the weekly schedule, graded pre-lab quizzes, submissions, and announcements are in [CATS](#). Worked solutions are released in CATS after the corresponding session.

Course documents

- [Course structure and instructional sequence](#)
- [Academic references and further reading](#)
- [From C to Silicon — toolchain, memory, and boot companion](#)

Lecture 1 - Embedded Systems, the C Toolchain, and Debugging

- [Theory slides](#)
- [Study sheet](#)
- [Practice worksheet](#)
- [Embedded C readiness diagnostic](#)
- Pre-lab quiz — in [CATS](#) under Tests & Quizzes (two attempts, highest score)
- [Lab guide](#)
- [Lab reference](#)

Lecture 2 — Values, Decisions, Loops, and Arrays

- [Theory slides](#)
- [Study sheet](#)
- [Practice worksheet](#)
- Pre-lab quiz — in [CATS](#) under Tests & Quizzes (two attempts, highest score)

- [Lab guide](#)

Lecture 3 — Functions, Arrays, Strings, and Multiple C Files

- [Theory slides](#)
- [Study sheet](#)
- [Practice worksheet](#)
- Pre-lab quiz — in [CATS](#) under Tests & Quizzes (two attempts, highest score)
- [Lab guide](#)
- [Starter files](#)

Lecture 4 — Pointers, Structures, Bits, Registers, and GPIO

- [Theory slides](#)
- [Study sheet](#)
- [Practice worksheet](#)
- Pre-lab quiz — in [CATS](#) under Tests & Quizzes (two attempts, highest score)
- [Lab guide — Bare-Metal GPIO](#)
- [Homework — External GPIO Control](#)

Lecture 5 — GPIO and Hardware Abstraction

- [Theory slides](#)
- [Study sheet](#)
- [Practice worksheet](#)
- Pre-lab quiz — in [CATS](#) under Tests & Quizzes (two attempts, highest score)
- [Lab guide — GPIO with the HAL](#)

Lecture 6 — Interrupts and the NVIC

- [Theory slides](#)
- [Study sheet](#)
- [Practice worksheet](#)
- Pre-lab quiz — in [CATS](#) under Tests & Quizzes (two attempts, highest score)
- [Lab guide — Button Interrupts \(EXTI\)](#)
- [Homework — HAL Interrupts](#)

Lecture 7 — Timers (TIM)

- [Theory slides](#)

- [Lab guide — Timer Interrupts](#)

Lecture 8 — Pulse-Width Modulation

- [Theory slides](#)
- [Lab guide — Fading LEDs and Controlling Servos](#)

Lecture 9 — Input Capture

- [Theory slides](#)
- [Lab guide — Measure Your Own PWM](#)

Lecture 10 — Analog-to-Digital Conversion (ADC)

- [Theory slides](#)
- [Lab guide — Reading the Analog World](#)

Lecture 11 — The Bridge to Assembly

- [Theory slides](#)
- [Lab guide — The Disassembly View](#)

Lecture 12 — Synchronous Serial: I2C and SPI

- [Theory slides](#)
- [Lab slides — Analog Sensors and LCDs](#)
- [Lab slides \(potentiometer version\)](#)

Lecture 13 — DMA, Displays, and Motors

- [Theory slides](#)
- [Lab slides — TFT Display and SPI with DMA](#)
- [DMA deck extract](#)

Lecture 14 — Final Project Presentations and Course Wrap-up

- [Theory slides](#)
- [Lab slides — Final Project Development and Demonstrations](#)